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THE ANALYSIS OF PROXIMITIES: MULTIDIMENSIONAL SCALING WITH AN UNKNOWN DISTANCE FUNCTION. I.

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A computer program is described that is designed to reconstruct the metric configuration of a set of points in Euclidean space on the basis of essentially nonmetric information about that configuration. A minimum set of Cartesian coordinates for the points is determined when the only available information specifies for each pair of those points—not the distance between them—but some unknown, fixed monotonic function of that distance. The program is proposed as a tool for reductively analyzing several types of psychological data, particularly measures of interstimulus similarity or confusability, by making explicit the multidimensional structure underlying such data.

Empirical procedures of several diverse kinds have this in common: they start with a fixed set of entities and determine, for every pair of these, a number reflecting how closely the two entities are related psychologically. The nature of the psychological relation depends upon the nature of the entities. If the entities are all stimuli or all responses, we are inclined to think of the relation as one of similarity. A somewhat more objective (though less intuitive) characterization of such a relation, perhaps, is that of substitutability. The statement that stimulus A is more similar to B than to C, for example, could be interpreted to say that the psychological (or behavioral) consequences are greater when C, rather than B, is substituted for A. From this standpoint a natural procedure for determining similarities of stimuli or responses is by recording substitution errors during identification learning [2, 7, 12, 14, 17, 18]. In addition, though, disjunctive reaction time and sorting time have also been proposed as measures of psychological similarity [20]. Finally, of course, individuals have sometimes been instructed simply to rate each pair of stimuli, directly, on a scale of apparent similarity [1, 6]. The notion of similarity is not necessarily restricted to stimuli or responses (in the narrow sense of these words), however. Serviceable measures of similarity may also be found for concepts, attitudes, personality structures, or even social institutions, political systems, and the like.

With some classes of entities, the notions of substitutability and, hence, similarity seem inappropriate. For example, data have been collected on the number of times each pair of persons in an isolated group communicates or otherwise associates with each other [3]. A number of this kind might be interpreted as a measure of the degree of association or mutual choice between two persons. It would not, however, be taken as a measure of their similarity. Experiments using word-association or free-recall methods furnish another example [16]. Although the word "butter" is frequently produced as an association to the word "bread," we are inclined to attribute this fact to an associative connection between the two words but not, again, to any high degree of similarity between these words or their referents.

Since the method of analysis to be described here may have some application to measures of association as well as to measures of similarity or substitutability, a generic term is needed to cover all of these various types of measures. Fortunately, there is one notion (albeit a rather abstract one) that does seem to run through all of these more specific concepts: that is the notion of psychological nearness, closeness, or degree of proximity. Thus one of two similar colors is said to be very *near* to the other, or two words are said to be *closely* associated. For all sets of data of this general kind, the number representing the closeness of the relation between a pair of entities will be called the *proximity measure* for that pair. The method of analysis for such numbers will accordingly be termed the *analysis of proximities.*

The notion of nearness or proximity, which is objectively defined only for pairs of objects in physical space, tends to be carried over to very different situations where the space in which entities can be closer together or further apart is not at all evident. As one possible basis for this extension of usage, Shepard [20] has argued that there is a rough isomorphism between the constraints that seem to govern all of these measures of similarity or association, on the one hand, and the metric axioms (which formalize some of the most fundamental properties of physical space), on the other. In particular, to the metric requirement that distance be symmetric, there is the corresponding intuition that if A is near B then B is also near A. To the metric requirement that the length of one side of a triangle cannot exceed the sum of the other two, there is the corresponding intuition that, if A is close to B and B to C, then A must be at least moderately close to C. The insertion of words like "very" and "moderately" attests to the loss of precision entailed by a shift from the operationally defined concept of distance to the intuitively defined notion of proximity. But this in turn invites an attempt to carry the powerful quantitative machinery that has developed around the concept of distance over into the qualitative area, where one has been able to speak only of nearness or proximity. An attempt of just this kind forms the subject matter of this paper.

If successful, such an attempt might advance us appreciably towards the solution of a problem that immediately confronts anyone who collects data of the proximity type; namely, the problem of the reduction of such data. The following example may help to indicate the magnitude of the reduction problem. There has been both theoretical and practical interest in the factors that determine confusions between morse code signals [10, 12, 13, 14, 15]. But, since there are 36 different signals to be investigated, at the very outset one is faced with the bewildering task of trying to find some sort of pattern or structure in the $N(N - 1)/2$ or 630 different similarities between these signals. Once some underlying pattern is found, though, the way would thereby be opened for the systematic exploration of how this pattern depends upon the physical properties of the stimuli, the stage of learning, prior training, the individual learner, and so on.

But the proposition has already been tendered that similarity is interpretable as a relation of proximity; this immediately suggests that the structure we are seeking in data of this kind is a *spatial* structure. Certainly we usually think that a greater degree of proximity implies a smaller distance of separation. If some monotonic transformation of the proximity measures could be found that would convert these implicit distances into explicit distances, then we should be in a position to recover the spatial structure contained only latently in the original data. Indeed, methods have already been developed whereby, given an explicit set of distances of this kind, one can map the stimuli, as points, into a Euclidean space (see Torgerson [21], pp. 254–259). Such a mapping constitutes a true reduction of the data since the originally bewildering array of proximity measures can then be reconstructed from a relatively much smaller set of coordinates for the resulting points in Euclidean space.

Clearly, the success of such an undertaking depends upon the selection of the proper distance function; that is, the function that will transform the proximity measures into Euclidean distances. Sometimes there is some independent information about the shape of this function. For example, Shepard [19] has adduced evidence for an approximately exponential relation between substitution errors during identification learning and the psychological distance between the stimuli or between the responses. Thus the distance function should presumably be logarithmic in form for this special case. Unfortunately, though, there is no reason to suppose that proximity measures obtained under quite different conditions will also be exponentially related to distance. Even in identification learning this relation apparently departs from the simple exponential as soon as feedback as to the correctness of the responses is curtailed [19]. As a matter of fact, in some cases we are primarily interested in discovering the shape of this unknown function, rather than the spatial configuration of the points.

Furthermore, even when the shape of the distance function is already known, there is this remaining difficulty: since proximity measures are usually bounded from below (typically by zero), they necessarily level off as distance becomes very large. The consequent nonlinearity of the distance function can magnify seemingly negligible statistical fluctuations of small proximity measures into wild swings of the corresponding estimates for the large distances. Because of this, stable configurations have been achieved in practice only by resorting to laborious smoothing procedures of a rather inelegant and ad hoc nature ([17], p. 341).

Basic Ideas for a New Approach

A quite different approach to this whole problem is described here—an approach that has become feasible, only recently, with the advent of digital computers of sufficient speed and capacity. In this approach, no assumption is made about the form of the distance function; the only requirement is that it be monotonic. The objectives of the analysis are to determine three things: (i) the minimum number of dimensions of the Euclidean space required such that the distances in this space are monotonically related to the initially given proximity measures, (ii) an actual set of orthogonal coordinates for the points in this minimum space, and (iii) a plot showing the true shape of the initially unknown function relating proximity to distance. It is a surprising outcome of the present research that the two conditions of monotonicity and minimum dimensionality (which seem nonmetric or qualitative in nature) are generally sufficient to lead to a unique and quantitative solution.

Achievement of Monotonicity

The first of the two fundamental conditions to be imposed on the solution, that of monotonicity, states that the rank order of the $N(N - 1)/2$ distances between the N points should be the inverse of the rank order of the $N(N - 1)/2$ proximity measures (at least to a sufficient degree of approximation). Thus the two points corresponding to the pair of entities with the largest proximity measure should end up with the smallest (or very nearly the smallest) distance of separation in Euclidean space. The device used to secure this condition of monotonicity is basically quite simple. First, any prescribed rank order of the distances whatever (including ties) can be realized so long as we are free to arrange the N points in a space of at least $N - 1$ dimensions. Suppose, then, that we have some particular arrangement of the points in an $(N - 1)$-dimensional space. If this particular configuration does not meet the requirement of monotonicity, then we should be able to find a better configuration simply by moving the points in such a way as to stretch those distances that are too small and compress those distances that are too large.

Stated more completely, the proposed procedure is as follows. First, from the $N(N - 1)$ coordinates specifying the initial configuration of points, the $N(N - 1)/2$ Euclidean distances between these points are computed. The computed distances are then rank ordered from smallest to largest, and the resulting ranking is compared with the ranking already obtained for the proximity measures. From each point $N - 1$ vectors are then constructed in such a way that each vector falls along the line connecting that point with one of the $N - 1$ other points. The vector is directed towards or away from the other point depending upon whether the rank of the distance between those two points is too large or too small (as determined from the rank of their proximity measure). The length of the vector is determined by the size of this discrepancy in rank. Thus the $N - 1$ vectors issuing from a point can be interpreted as $N - 1$ forces that are tending to pull that point towards those other points that are too distant and away from those other points that are too close.

Once the set of $N - 1$ vectors has been determined for all N points, these points are simultaneously displaced in their $(N - 1)$-dimensional space. The vector governing the displacement of each point is computed as the vector sum (or resultant) of the $N - 1$ individual vectors attached to that point. Since the displacement of any one point is a compromise between $N - 1$ separate tendencies, there is no guarantee that the simultaneous displacement of all N points will immediately instate the desired condition of inverse ranking. But there is no need to insist that the final solution be achieved in a single stroke. Rather, the procedure can be iterated, each time on the basis of the successively smaller residue of departures from monotonicity remaining from all preceding iterations.

In order to get the iterative process under way, some starting coordinates must be supplied. Fortunately coordinates can always be found in $N - 1$ dimensions such that all $N(N - 1)/2$ interpoint distances are equal. This insures that no bias will be introduced into the final solution. The points having this property coincide with the vertices of a regular simplex. (This configuration is simply the generalization, to higher spaces, of the equilateral triangle in two dimensions and the regular tetrahedron in three dimensions.) As the iterative process proceeds, then, each point moves away from its initial position at one vertex of the regular simplex. The trajectory of the point is controlled by vector forces that are constantly changing so as to satisfy the requirement of monotonicity. The whole configuration comes to rest only after the rank order of the distances is the inverse of the rank order of the initially given proximity measures.

Achievement of Minimum Dimensionality

Actually the process, as just described, would come to rest almost immediately. This is because a trivial solution can always be found in $N - 1$ dimensions by making arbitrarily small adjustments in the regular simplex. An intuitive feeling for this fact can be gained by considering the special case of four points in ordinary three-space. Initially these four points will coincide with the vertices of a regular tetrahedron (a figure that cannot be isometrically embedded in a space of fewer than three dimensions). Clearly, any one of the six edges of this tetrahedron could now be shortened by some small amount Δs without affecting the lengths of the other five edges. (It is sufficient to subject two of the four triangular faces to a very small rigid rotation, towards each other, about the edge opposite the one to be shortened.) Moreover, any one of the other five edges could then be shortened by some amount smaller than Δs in exactly the same way without altering any of the other edges.

This process could be continued so that each edge (taken in any sequence) is shortened by a successively smaller amount. The final result would be a tetrahedral configuration in which the distances between the four vertices conform to any predesignated ordering. The extension of this argument to the general case of N points in $N - 1$ dimensions need not be elaborated here; for the fact that all orderings of distances are possible in $N - 1$ dimensions follows from results already presented by Bennett and Hays ([4], pp. 37–38). Taken by itself, then, the method outlined above for securing monotonicity is of little utility. It leads to no real reduction of the original data. Furthermore, it does not yield a unique solution, since the amount each successive edge is to be shortened is not completely specified—it can, in fact, be chosen in infinitely many ways.

A nontrivial and unique solution can only be secured by supplementing the monotonicity requirement with an additional requirement; namely, the final configuration should be of the smallest possible dimensionality. As a helpful heuristic, consider a mapping of a linear segment of length s onto a semicircular segment spanning θ radians of a circle. The mapping can be chosen so that the distance d between any two points of the segment is transformed into the new distance $f(d)$ given by

$$f(d) = r\sqrt{2[1 - \cos(\theta\, d/s)]} \qquad (0 < \theta \le \pi). \tag{1}$$

Here r denotes the radius of curvature of the semicircular segment. This radius can always be chosen so that the mean interpoint distance remains invariant during the mapping. For example, if the N points are evenly distributed along the segment, invariance is approached (for large N) by choosing

$$r = \frac{s\theta}{24[1 - (2/\theta)\sin(\theta/2)]},$$

in which case the mean distance remains fixed in the vicinity of $s/3$.

In any case, since $0 < \theta \le \pi$, the function f is monotonic; hence the rank order of the interpoint distances is necessarily preserved. Clearly, then, a choice cannot be made between the straight and the curved segments on the basis of the rank order of the distances alone. Note, however, that whereas the reconstruction of the distances from Cartesian coordinates can be effected with only one coordinate axis in the case of the linear segment, two such axes are required for the semicircular segment. Note, also, that whereas there is only one linear configuration, there is a different semicircular configuration for each choice of the parameter θ governing the total curvature of the segment. Thus considerations of simplicity or parsimony, on the one hand, and of uniqueness, on the other, lead us to prefer the solution that can be represented in the space of lesser dimensionality.

In this particular example the inverse of the function f can be used to flatten the semicircular segment back into the straight line by systematically stretching the larger distances and shrinking the smaller distances. A number of other examples reinforce the inference that, in general, the collapsing of a configuration into a space of smaller dimensionality is accompanied by an increase in the variance of the distances. Suppose, in particular, that a set of points is to be confined to a hyperspherical region of fixed radius. We have already noted that the case of minimum variance of the interpoint distances can only be achieved in $N - 1$ dimensions, where the vertices of a regular simplex are all separated by exactly the same distance. At the other extreme, if the rank order of the distances is disregarded, then the case of maximum variance can only be realized by dividing the points evenly between two spatially separated clusters in such a way that the distances between any two points within the same cluster approaches zero. This is clearly a one-dimensional configuration, since all distances can be reconstructed from coordinates on a single axis passing through the two clusters. More generally, Hammersley [8] has shown that the variance of the distribution of distances in a hyperspherical region of K-dimensional space is a decreasing function of the number of dimensions K. Indeed, this distribution is asymptotically normal with variance decreasing approximately as the reciprocal of the number of dimensions [8, 11].

These considerations suggest that the way to flatten a configuration into a space of smaller dimensionality is to increase the variance of the distances by further stretching those distances that are already large and by further shrinking those distances that are already small. This is just what is done in the method of analysis proposed here. During each iteration two sets of $N - 1$ vectors are computed for each of the N points. While the first set (already described) is designed to improve the position of the point with respect to the requirement of monotonicity, the second set is intended to move that same point closer to nearby points and further from remote points. Since the proximity measures are, by hypothesis, monotonically related to the distances, these measures can be used in place of the distances themselves to compute the second set of vectors. (This device seems to avoid some instabilities inherent in the other alternative.) In particular, for any one point a vector of this second type is extended towards or away from each other point depending upon whether the proximity measure for the two points is larger or smaller than the mean for all $N(N - 1)/2$ proximity measures. The length of the vector, again, is proportional to the magnitude of this deviation from the mean.

Hence the final displacement of each point during one iteration is the resultant of altogether $2(N - 1)$ vectors. Half of these represent forces that are acting to achieve monotonicity; the remaining half represent forces that are acting to drive the configuration into a space of smaller dimensionality. With the addition of this second set of vectors, therefore, the spatial configuration continues to change until both criteria are satisfied. At that point there are no further changes, since the vectors tending to flatten the configuration still further are exactly counteracted by the vectors tending to maintain monotonicity.

The convergence to this final configuration does not immediately supply the reduction we are seeking, however, since each of the N points is still specified by $N - 1$ coordinates. The situation is analogous to a coplanar set of points in ordinary three-space. In order to reduce the number of coordinates for each point from three to two, in that case, the coordinate system (with origin at the centroid) must be rotated so that all projections on the third axis vanish. Only then can the distances between the coplanar points be reconstructed from their coordinates on two axes alone. Similarly in the general case, the attainment of the most economical description requires that the coordinate system be rotated into "principal axes." When this has been done the total variance of the coordinates (which is invariant under rotation) will be partitioned among the axes in such a way that each successive axis accounts for the greatest possible fraction of that variance not accounted for by all preceding axes. When only a negligible fraction remains to be accounted for, all succeeding axes can be eliminated as superfluous. Hopefully, the number of axes that must be retained will be small compared with the number of points N. For it is the set of coordinates on these retained axes that constitutes the reduced characterization of those points and, hence, of the original proximity measures.

Method of Computation

A program for carrying out the operations outlined only roughly in the preceding sections has now been compiled and tested on an IBM 7090 computer. The remainder of this paper is devoted to the presentation of a more complete and precise statement of the mathematical details of this program.* A second paper (Part II, to follow in a subsequent issue of this journal) will present the results of a number of tests that were performed to show that this program does in fact yield solutions of the kind claimed for it.

Preparations for the Iterative Procedure

As soon as the $N(N - 1)/2$ initially given proximity measures are fed into the computer, two operations are performed. First, these measures are linearly transformed to bring them into a standard form in which the largest

*Copies of the actual FORTRAN listing for this program will be made available, by the author, to those who might wish to use it.

measure is unity and the smallest measure is zero. The standardized proximity measure obtained in this way for the points i and j will be denoted by s_{ij}. Second, these standardized measures are then rank ordered from smallest to largest. Tied measures are, for convenience, put into an arbitrary order. (Owing to a refinement in the use of the ranking, this treatment of ties has no influence upon the outcome.)

The next step is to generate a set of coordinates for each of the N vertices of a regular simplex in $N - 1$ dimensions. These are the starting coordinates that will be modified during the first iteration. The orientation of the simplex is arbitrary but, again for convenience, its position and size are so chosen that the centroid is at the origin and all edges are of unit length. Specifically, if x_{ia} denotes the coordinate for vertex i on axis a, then

$$x_{i(2q-1)} = \frac{\cos [2q(i - 1)\pi/N]}{\sqrt{N}}, \tag{2a}$$

$$x_{i(2q)} = \frac{\sin [2q(i - 1)\pi/N]}{\sqrt{N}}, \tag{2b}$$

as q runs from one to the greatest integer in $(N - 1)/2$. In case N is even, the coordinates on the last axis are given by

$$x_{i(N-1)} = \frac{(-1)^{i-1}}{\sqrt{2N}}. \tag{2c}$$

That the N points with these coordinates are in fact all separated by unit distances can easily be shown (see Coxeter [5], p. 245).

Comparison of Rank Orders

At the beginning of each iteration, the Euclidean distances between the N points are calculated from their $(N - 1)$-dimensional coordinates for that iteration by means of the generalized Pythagorean formula

$$d_{ij} = \sqrt{\sum_{a=1}^{N-1} (x_{ia} - x_{ja})^2}. \tag{3}$$

These distances are then ranked from smallest to largest for purposes of comparison with the ranking already determined for the proximity measures. Tied distances are always put into an order that is exactly the opposite of the order dictated by the ranking of the proximity measures. This is done to insure that distances that should not be tied will be pulled apart during the iteration.

A certain refinement is introduced into the comparison of the two rankings for the following reason. Although any ranking of the distances can be exactly realized in $N - 1$ dimensions, some small violations of monotonicity must be anticipated whenever the configuration is forced into a space of smaller dimensionality—particularly since the initial proximity measures are usually subject to errors of measurement. However, some departures from perfect inverse ranking can be tolerated better than others. Consider, for example, a case in which the violation of monotonicity consists merely in the reversal of two distances of adjacent ranks. Clearly, the seriousness of such a reversal would depend upon the magnitude of the difference between the two corresponding proximity measures. Indeed, if these two proximity measures were sufficiently close together, the direction of their difference would not be statistically reliable anyway. In that event, the reversal of the ranks of the two distances could be disregarded. In comparing ranks, discrepancies should somehow be weighted to reflect the magnitude of the difference between the proximity measures.

Actually such a refinement is quite easy to implement. Instead of comparing the ranks themselves, these integers are first replaced by the corresponding standardized proximity measures. To be more precise, let $R(s_{ij})$ denote the rank of the proximity measure s_{ij} for the two points i and j; let $R'(d_{ij})$ denote the *reversed* rank of their distances so that, for perfect inverse ranking, $R(s_{ij}) = R'(d_{ij})$. An obvious measure of the departure from monotonicity of the rank of the distance between i and j is the algebraic difference $R(s_{ij}) - R'(d_{ij})$. However, in place of this, the present proposal is to use the more refined measure $s_{ij} - s(d_{ij})$, where $s(d_{ij})$ stands for the proximity measure of rank $R'(d_{ij})$. This amounts, in effect, to a comparison of each proximity measure with its corresponding distance after the scale of distance has been subjected to a nonlinear transformation that renders the distribution of distances identical with the distribution of proximity measures. If the relation between these two variables is monotonic, then this procedure insures that it will also be linear.

Vectors for the Approach to Monotonicity

Since the standardized proximity measures range from zero to one, the algebraic differences $s_{ij} - s(d_{ij})$ can assume any value between plus and minus unity. A positive value indicates that the distance is too great, a negative value that the distance is too small, and zero that there is no departure from monotonicity for that pair of points. The vectors designed to achieve a closer approach to monotonicity during a given iteration can therefore be constructed on the basis of these algebraic differences. In particular, the component on axis a for the vector directed from point i along the line passing through some other point j is given by

$$\alpha_{ija} = \frac{\alpha[s_{ij} - s(d_{ij})](x_{ja} - x_{ia})}{d_{ij}}, \tag{4}$$

where the parameter α (without subscripts) is a constant multiplier for the lengths of all vectors. Larger values of α promote more rapid convergence. But, as in all error-correcting systems of this kind, if α is chosen too large the system over-corrects and, hence, degenerates into nonconvergent oscillation.

Vectors for the Approach to Minimum Dimensionality

A second set of vectors is then constructed in order to induce a slight stretching of the larger distances and shrinkage of the smaller distances. The component on axis a for the vector (of this second type) directed from point i along the line passing through point j is given by

$$\beta_{ija} = \frac{\beta[s_{ij} - \bar{s}](x_{ja} - x_{ia})}{d_{ij}}, \tag{5}$$

where $\bar{s}$ is the mean of all $N(N - 1)/2$ proximity measures. Again the parameter β (without subscripts) governs the rate at which the configuration is forced into spaces of smaller and smaller dimensionality. As will be seen β must generally be less than α. Otherwise the tendency to flatten the configuration overrides the tendency to maintain monotonicity, and the configuration is consequently driven into a space of spuriously low dimensionality.

Displacement of the Points

The vector specifying the actual displacement of point i during the given iteration is computed as the vector sum of both types of vectors. Explicitly, the component of the displacement on axis a is given by

$$\Delta x_{ia} = \sum_j (\alpha_{ija} + \beta_{ija}), \tag{6}$$

where the summation is carried out over all $N - 1$ other points $j(\neq i)$. Thus, during one iteration the coordinate for point i on axis a is changed from x_{ia}, at the beginning of the iteration, into the new value x'_{ia} given by

$$x'_{ia} = x_{ia} + \Delta x_{ia}. \tag{7}$$

Finally, at the end of each iteration the adjusted coordinates x'_{ia} undergo a "similarity" transformation to re-center the centroid at the origin and to re-scale all distances so that the mean interpoint separation is maintained at unity. The adjusted and transformed coordinates are treated in exactly the same way during the following iteration. The iterative process can be terminated when the adjustments Δx_{ia} have become too small to have any appreciable effect on the configuration.

Rotation to Principal Axes

Following the last iteration, the coordinate system should be rotated to principal axes. To this end, the matrix of scalar products of vectors from the centroid to all pairs of points is constructed. The scalar product for the points i and j is given by

$$b_{ij} = \sum_{a=1}^{N-1} x_{ia}x_{ja}. \tag{8}$$

Next, the characteristic roots and vectors of this matrix (which is real and symmetric) are computed by the Jacobi method of diagonalization (e.g., [9], pp. 180–185). These roots, with their associated vectors, are then ordered so that the roots form a decreasing sequence. The ath such root λ_a specifies the variance accounted for by the ath most important of the rotated axes. If ξ_{ia} denotes the ith component of the characteristic vector associated with λ_a, then the actual coordinates on the ath rotated axis are given by

$$x_{ia}^* = \xi_{ia}\sqrt{\lambda_a}\,. \tag{9}$$

Hopefully the first few axes will account for almost all of the variance. The coordinates on the remaining axes can then be eliminated to achieve the reduced representation of the spatial configuration.

After rotation to principal axes it is sometimes desirable to carry out further iterations in the collapsed space. The reason for this is as follows. First, as long as $\beta > 0$ (as it must be to achieve minimum dimensionality) some slight distortion is necessarily introduced into the stabilized configuration. This distortion can be reduced to any desired level by choosing β sufficiently small. But this, in turn, leads to a slower convergence. In order to conserve computing time, therefore, it is often expedient to select a larger β and even to rotate to principal axes before the configuration has come completely to rest.

The results of this premature rotation cannot of course be taken as a final solution. These results can be used to make an inference as to the number of dimensions required and to provide a set of approximate coordinates in the space of this number of dimensions. This approximate solution can then be brought into its exact form by setting $\beta = 0$ and iterating to a final configuration in the collapsed space. Since there will be no stretching or shrinking with $\beta = 0$, the only criterion governing the final configuration is that of monotonicity. If a sufficient degree of monotonicity is not achieved, this indicates that the number of required dimensions has been underestimated. In this event one can easily return to the coordinates obtained from the "premature" rotation, add the next axis, and then iterate to a new solution in the resulting augmented space.

Criterion for Terminating the Iterative Process

An explicit measure of the over-all departure from monotonicity remaining after any iteration is the mean square discrepancy between the adjusted rank numbers; namely,

$$\delta = \frac{2\sum_{j=2}^{N}\sum_{i=1}^{j-1}[s_{ij} - s(d_{ij})]^2}{N(N-1)}. \tag{10}$$

In the following section and in the sequel (Part II) δ will be shown to provide a satisfactory indicator for determining when to perform the "premature" rotation as well as when to terminate, finally, the iterative process.

Choice of Values for α and β

Owing to the trading relation, already noted, between speed and accuracy in the process of convergence, it seemed desirable to undertake a systematic exploration of the effects of these two parameters. In order to provide for an objective evaluation of convergence, a set of artificial proximity measures was used. These were generated by applying a known function to the distances between points in a known configuration. Thus the spatial configuration reconstructed during any stage of the iterative process could be directly compared with the known "true" configuration. In order to save computing time, a two-dimensional array of only five points was selected as this known configuration. For convenience, the scale of distance was chosen so that the mean of the ten interpoint distances was unity. The set of ten artificial proximity measures was then generated by arbitrarily defining the proximity measure for any two points to be a normal (Gaussian) function of the distance between them.

The long rectangular panel in the upper left corner of Fig. 1 shows how the discrepancy between the configuration reconstructed during each iteration and the "true" configuration of five points declined during 80 consecutive iterations with $\alpha = .20$ and $\beta = .05$. The measure of discrepancy used is the sum of squared deviations of the ten reconstructed distances from the ten true distances divided by the sum of squared deviations of the true distances from their mean (which in this case is unity). Since the ten reconstructed distances are initially all unity, this measure of discrepancy necessarily starts at 1.0 before the first iteration. Note that, with these values for α and β, there was a very marked drop in the discrepancy during the first iteration; convergence to the true solution was essentially achieved by the 15th iteration. Thereafter, the discrepancy remained below .02.

The triangular array of square panels in the lower right section of the figure uses the same measure of discrepancy to exhibit the analogous curves for 21 combinations of values of α and β. The 32-fold range from 0.05 to 1.60 is covered for both parameters. But each panel displays the results for the first 20 iterations only. (Thus the panel for $\alpha = .20$ and $\beta = .05$ in this array duplicates the square section at the left of the upper rectangular graph already described.) Apparently, for small values of α, i.e., to the left in the two-dimensional array of panels, the curves tend to be smooth and orderly; whereas for large values of α, i.e., to the right, the curves tend to be more jagged and erratic. Indeed, the excursions for $\alpha = 1.60$ were so large that only the occasional re-entries into the panels could be shown.

Apart from these oscillations for large values of α, the curves tend to decrease more or less monotonically for $\beta < \alpha/4$, say. But, for larger values

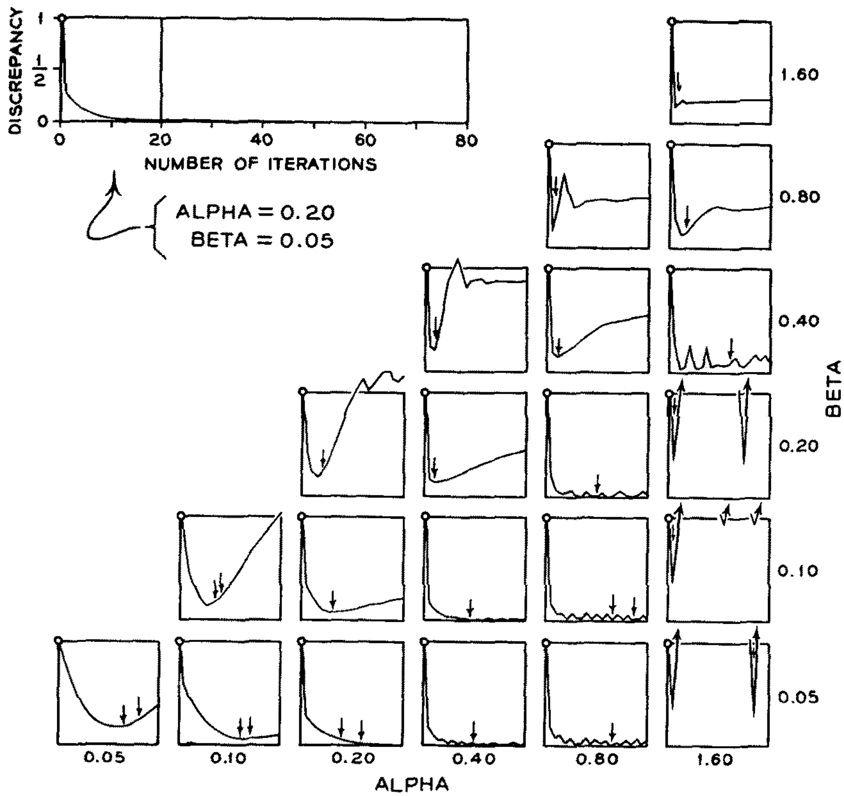

FIGURE 1

Discrepancy between the reconstructed configuration and the true configuration as a function of number of iterations for each of 21 different combinations of values for α and β.

of β, the curves pass through a minimum and then increase again. Evidently the most precise convergence is achieved by choosing α as small as .20 and by choosing β considerably smaller than α. Thus, if computing time is of little concern, a solution of high accuracy could probably be achieved with $\alpha = .20$ and $\beta = .01$.

Unfortunately, when N is at all large convergence can be quite slow with these values for α and β; hence, computing time does become a matter for concern. The curves in Fig. 1 suggest a more expedient strategy in such cases. Note, for example, the curve for $\alpha = .4$ and $\beta = .2$. Although this curve rises after the second iteration, it does indicate a rather close approach to the true configuration after only two iterations. Other neighboring curves exhibit similar behavior. If there were some way to establish just when the discrepancy is minimum, the coordinate system could be rotated to principal axes at that point. The final solution could then be secured rather quickly by iterating in the collapsed space with $\alpha = .2$ and $\beta = .0$, say. Of course, since the true configuration is not generally known in advance, curves like those shown in Fig. 1 cannot be plotted in practice. However, the measure of the over-all departure from monotonicity, δ, can always be computed. Moreover, this measure turns out, in general, to attain its minimum very close to the point at which the discrepancy between the distances is minimum.

The points at which the quantity δ is minimum are designated by the small vertical arrows above the curves in Fig. 1. (When there are two arrows, these indicate the first and last iterations for which the departure from monotonicity attained its minimum value during the twenty iterations. When N is as large as ten or fifteen, however, this departure usually reaches its minimum only once during a comparable number of iterations.) These results suggest—and subsequent experience confirms—that the number of dimensions can be adequately estimated on the basis of the variances of the coordinates on the principal axes when the departure from monotonicity is minimum. The effectiveness of this strategy, particularly for cases of substantially larger N, will become apparent in the sequel (Part II). That paper will be devoted to the detailed description and evaluation of a number of test applications of the method presented here.

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